UP-VLA: A Unified Understanding and Prediction Model for Embodied Agent

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Abstract

Vision-Language-Action (VLA) models inherit rich semantic knowledge from large pre-trained vision-language models but often struggle with the low-level spatial cues and physical dynamics essential for precise robotic manipulation. Existing pre-training paradigms focus on high-level multi-modal understanding while neglecting predictive, dynamics-aware objectives, which limits both precision and generalization in downstream manipulation tasks. In this work, we present **UP-VLA**, a unified VLA framework that jointly optimizes multi-modal understanding (MMU) and future visual prediction (PRE) within a single autoregressive transformer backbone. Built on the Show-o architecture with Phi-1.5 as the language model, UP-VLA employs a dual visual stream: a continuous CLIP-ViT encoder for semantic understanding and a discrete VQ-GAN tokenizer for future image generation. A task-aware prompting and attention masking scheme enables the model to seamlessly switch between understanding, prediction, and action prediction modes. Training proceeds in two stages: first co-pretraining MMU on the LLaVA-665k dataset and PRE on the Bridge dataset, then jointly fine-tuning on downstream robot demonstrations while continuing MMU co-training. Extensive experiments demonstrate that UP-VLA achieves state-of-the-art performance on the challenging CALVIN long-horizon benchmarks, obtaining an average task completion length of 4.08 on ABC→D (+33% over GR-1) and 4.42 on ABCD→D. Real-world Panda robot experiments further confirm strong generalization, achieving 0.58 success rate on unseen objects—a 190% relative improvement over the model variant without the MMU component. These results highlight the complementary benefits of semantic understanding and future prediction for embodied agents.

1. Introduction

Building robots that can understand natural language instructions and execute complex manipulation tasks in unstructured environments is a central challenge in embodied AI. Recent advances in large-scale pre-trained vision-language models (VLMs) have inspired a new class of Vision-Language-Action (VLA) models (Brohan et al., 2023; Kim et al., 2024; Black et al., 2024; Li et al., 2023b) that leverage rich semantic knowledge from internet-scale data. By initializing from pre-trained VLMs and fine-tuning on robot demonstrations, these approaches achieve impressive instruction-following capabilities and cross-task generalization.

However, despite their semantic strengths, VLA models often fall short in tasks requiring precise spatial understanding and awareness of physical dynamics. For instance, a robot may correctly parse an instruction such as "pick up the red block and place it on the tray" yet fail to execute the precise reaching trajectory due to insufficient spatial grounding. This limitation stems from a fundamental mismatch: VLM pre-training objectives (e.g., visual question answering, image captioning) are designed for high-level semantic alignment and do not directly encourage the model to reason about low-level geometric relationships or anticipate how a scene will evolve after an action is taken.

An orthogonal line of work addresses this gap through *predictive* pre-training. Models such as GR-1 (Wu et al., 2023) and UniPi (Du et al., 2024) train policies to generate future image frames alongside robot actions, which forces the model to internalize physical dynamics and scene geometry. These approaches have demonstrated strong generalization on long-horizon benchmarks (Mees et al., 2022). Yet they largely forgo the rich semantic grounding provided by large-scale language-vision pre-training, limiting their instruction comprehension and transfer to novel object categories.

We argue that **semantic understanding** and **future visual prediction** are complementary objectives for embodied agents: understanding enables high-level reasoning and instruction grounding, while prediction captures low-level spatial details and physical dynamics. This motivates **UP-VLA** (**U**nified understanding and **P**rediction **VLA**), a model that trains both objectives jointly within a single autoregressive transformer.

Concretely, UP-VLA is built on the Show-o architecture (Xie et al., 2024), a unified transformer backbone (Phi-1.5 (Li et al., 2023c)) that supports both multi-modal understanding and discrete image generation. We extend this backbone for robotic control by introducing: (1) a *dual visual stream* with a continuous CLIP-ViT (Radford et al., 2021) encoder for understanding and a VQ-GAN tokenizer (Esser et al., 2021; Yu et al., 2023) for discrete future image prediction; (2) a *task-aware prompting and attention* scheme that places image tokens before language for understanding (MMU mode) or after language for prediction (PRE mode), controlled by a special `[PRE]` token; and (3) a *two-stage training pipeline* that first co-pretrains on internet-scale VQA data (LLaVA-665k) and robot video data (Bridge), then fine-tunes on downstream demonstrations while jointly predicting future images and multi-step actions.

Our main contributions are:

- We introduce UP-VLA, a unified VLA training paradigm that co-optimizes multi-modal understanding and future visual prediction to capture both semantic knowledge and low-level spatial dynamics within a single autoregressive backbone.
- We design a task-aware prompting and attention mechanism that enables the model to seamlessly handle understanding, prediction, and action learning modes with no architectural overhead.
- We achieve state-of-the-art performance on CALVIN long-horizon benchmarks and demonstrate strong generalization to unseen objects in real-world manipulation experiments.
- We provide comprehensive ablations showing that the MMU and prediction pre-training components contribute complementary benefits, and that removing either substantially degrades performance.

2. Related Work

Vision-Language-Action Models. VLA models leverage pre-trained vision-language foundations for robotic control. RT-2 (Brohan et al., 2023) fine-tunes a large VLM (PaLI-X) to directly output tokenized actions, transferring web-scale knowledge to manipulation. OpenVLA (Kim et al., 2024) provides an open-source VLA trained on the Open X-Embodiment dataset (O'Neill et al., 2023). RoboFlamingo (Li et al., 2023b) adapts OpenFlamingo with a dedicated policy head for sequential decision-making, demonstrating strong results on the CALVIN benchmark. PaLM-E (Driess et al., 2023) integrates continuous sensor modalities directly into a large language model for embodied planning. π_0 (Black et al., 2024) employs flow matching on top of a pre-trained VLM for dexterous manipulation. 3D-VLA (Zhen et al., 2024) augments VLA with 3D scene representations and a generative world model. HIRT (Zhang et al., 2024) and TraceVLA (Zheng et al., 2024) explore hierarchical and trace-based control structures for VLA models. While these approaches excel at semantic instruction grounding, they typically do not explicitly model physical dynamics or future scene evolution.

Prediction-Based Robot Learning. A complementary direction learns policies by predicting future observations. UniPi (Du et al., 2024) formulates policy learning as text-conditioned video generation, using a diffusion model to plan in video space. SuSIE (Black et al., 2023) leverages a pre-trained image-editing diffusion model to generate subgoal images, guiding a low-level goal-conditioned policy. GR-1 (Wu et al., 2023) is a GPT-style transformer that simultaneously predicts future frames and robot actions after pre-training on large-scale video data, achieving strong performance on CALVIN. PAD (Guo et al., 2024) unifies image prediction and action learning in a joint diffusion process. Diffusion Policy (Chi et al., 2023) and 3D Diffuser Actor (Ke et al., 2024) use diffusion-based action generation with visuomotor and 3D scene representations. Unlike these methods, UP-VLA additionally incorporates large-scale language-image understanding pre-training, enabling stronger semantic grounding alongside dynamics prediction.

Unified Multimodal Models. There is growing interest in unified transformers that handle both multimodal understanding and generation (Wang et al., 2022; Ge et al., 2024; Dong et al., 2023). Show-o (Xie et al., 2024) demonstrates that a single transformer can unify autoregressive text-image understanding with discrete diffusion image generation. BLIP-2 (Li et al., 2023a) and InstructBLIP (Dai et al., 2024) bootstrap VQA capabilities from frozen encoders. LLaVA (Liu et al., 2024) achieves strong instruction-following for vision-language tasks. UP-VLA draws inspiration from these unified models, extending the Show-o paradigm to the embodied domain by incorporating robot action prediction and physical prediction objectives.

World Models for Embodied Agents. World models that predict future states have a rich history in model-based reinforcement learning. Extending this idea to language-conditioned robotics, future image prediction serves a dual purpose: it provides training signal for the model to internalize physical dynamics without requiring explicit action labels, and it can guide action selection by anticipating the consequences of candidate motions. Prior prediction-based policies such as GR-1 (Wu et al., 2023) and UniPi (Du et al., 2024) exploit this for improving spatial precision, but lack rich semantic grounding. UP-VLA bridges both paradigms, conditioning action generation on predicted future observations while simultaneously maintaining VQA-level semantic

understanding—making it uniquely suited to tasks that require both precise spatial reasoning and flexible language grounding.

Self-Supervised Pre-training for Robot Learning. A complementary research direction develops self-supervised visual representations tailored to robotic control. R3M (Nair et al., 2022) pre-trains a visual encoder on large-scale human video via time-contrastive and language alignment objectives, yielding representations that transfer effectively to manipulation tasks with limited robot demonstrations. LIV (Ma et al., 2023) learns joint language-image value functions from action-free annotated videos, enabling dense reward specification without environment interaction. While these methods produce compact, task-relevant visual encoders, they typically treat the pre-trained backbone as a frozen feature extractor separate from the policy network. In contrast, UP-VLA integrates understanding pre-training directly into the policy backbone in an end-to-end manner, allowing all components to co-adapt during robot fine-tuning and ensuring that semantic representations remain responsive to the manipulation data distribution.

3. Preliminaries

Show-o Backbone. UP-VLA is built upon the Show-o (Xie et al., 2024) unified transformer, which uses Phi-1.5 (Li et al., 2023c) as its language model backbone. Show-o supports two visual modalities: (1) continuous image features from a CLIP-ViT (Radford et al., 2021) encoder for multi-modal understanding tasks such as VQA, and (2) discrete image tokens from a VQ-GAN / MAGViT-v2 (Esser et al., 2021; Yu et al., 2023) tokenizer for autoregressive image generation. The model processes sequences of text tokens and visual tokens jointly through a causal transformer, with different attention patterns for the two modes.

Problem Formulation. We consider language-conditioned robot manipulation. At each timestep t, the robot receives an RGB observation $o_t \in \mathbb{R}^{H\times W\times 3}$, a language instruction ℓ, and optionally a history of past observations $\{o_{t-k}, \dots, o_{t-1}\}$. The goal is to predict a sequence of future actions $\{a_t, a_{t+1}, \dots, a_{t+T-1}\}$ that accomplish the instruction. We additionally augment this with a future image prediction task: predicting the next observation $\hat{o}_{t+1}$ conditioned on the current state and the action to be taken. This dual objective encourages the policy to reason explicitly about scene dynamics.

VQ-GAN Tokenization. For future image prediction, we discretize images into a sequence of tokens using a VQ-GAN-style encoder (Esser et al., 2021). Given an image $o \in \mathbb{R}^{H\times W\times 3}$, the tokenizer maps it to a grid of discrete codes $\mathbf{z} = (z_1, z_2, \dots, z_N) \in \mathcal{V}^N$, where $\mathcal{V}$ is the codebook vocabulary and $N = (H/p) \times (W/p)$ for patch size p. The decoder reconstructs the image from these codes. This representation enables autoregressive prediction of future images as a sequence of discrete tokens, which can be handled by the same transformer backbone as language tokens.

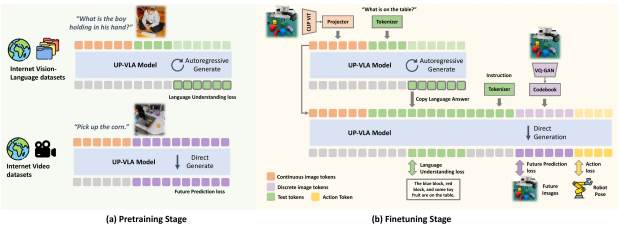

Figure 1. Overview of UP-VLA. The model uses a dual visual stream: continuous CLIP-ViT features for multi-modal understanding (MMU) and discrete VQ-GAN tokens for future image prediction (PRE). A shared autoregressive transformer (Phi-1.5) processes all inputs and jointly predicts actions, future images, and text responses.

4. Methodology

4.1. Overall Architecture

Figure 1 provides an overview of UP-VLA. The model consists of three main components: (1) a CLIP-ViT visual encoder for continuous semantic features used in MMU, (2) a VQ-GAN tokenizer for discrete image tokens used in future prediction, and (3) the shared Phi-1.5 autoregressive transformer that processes all token sequences and outputs actions, future image tokens, and text responses.

4.2. Dual Visual Stream

Multi-Modal Understanding Stream (MMU). For the MMU stream, we use a CLIP-ViT encoder (Radford et al., 2021) to extract continuous patch-level features from the current observation o_t. These features are projected to the transformer embedding dimension via a linear projection layer, yielding a sequence of visual tokens $\mathbf{v}^{\text{mmu}} = (v_1, \dots, v_M)$, where M is the number of image patches. These tokens are used for tasks requiring rich semantic alignment, such as VQA and scene description generation.

Future Prediction Stream (PRE). For the PRE stream, we encode both the current observation o_t and the predicted next observation $\hat{o}_{t+1}$ into discrete token sequences using the VQ-GAN tokenizer. The current observation is tokenized as condition tokens, and the model autoregressively generates the future image tokens. This discrete tokenization allows future image prediction to be formulated as a standard next-token prediction problem, compatible with

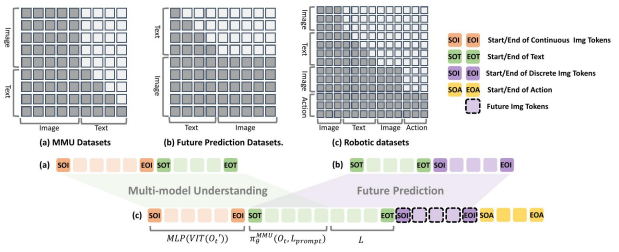

Figure 2. Illustration of the unified prompting and attention mechanism in UP-VLA. For MMU tasks, image tokens appear before language tokens with bidirectional attention on image tokens. For PRE tasks, language tokens appear before future image tokens, and a special `[PRE]` token triggers the prediction mode.

the language model objective.

4.3. Task-Aware Prompting and Attention

The key challenge in unifying understanding and prediction within a single model is managing the different token orderings required by each task. We adopt a *task-aware prompting and attention* mechanism, illustrated in Figure 2, inspired by Show-o (Xie et al., 2024).

MMU Mode. For multi-modal understanding tasks, the input sequence is arranged as:

$$[\text{sys}]\ \mathbf{v}^{\text{mmu}}\ [\text{Q}]\ \mathbf{t}_{\text{question}}\ [\text{A}], \tag{1}$$

where [sys] is the system prompt, $\mathbf{v}^{\text{mmu}}$ are the CLIP-ViT visual tokens placed *before* the text question, and [A] precedes the generated answer tokens. The image tokens attend to each other bidirectionally, while subsequent tokens attend causally. This ordering encourages the model to ground its language responses in the visual context.

PRE Mode. For future image prediction, the input sequence is:

$$[\text{sys}]\ \mathbf{v}^{\text{img}}\ [\text{Q}]\ \mathbf{t}_{\text{inst}}\ [\text{PRE}]\ \hat{\mathbf{z}}, \tag{2}$$

where $\mathbf{v}^{\text{img}}$ are the current observation tokens (from VQ-GAN), $\mathbf{t}_{\text{inst}}$ is the language instruction, and [PRE] is a learned special token that signals the model to enter prediction mode and generate the future image token sequence $\hat{\mathbf{z}}$. The causal attention mask applies to all tokens, so the future tokens can only attend to past tokens. This formulation allows the model to condition future predictions on both the visual state and the language goal.

Action Mode. During robot execution, UP-VLA generates actions by conditioning on the current observation, instruction, and a self-generated scene description from the MMU module. The MMU module first produces a textual description of the current scene (e.g., object positions, grasping state), which is then appended to the action prediction context. This *MMU-conditioning* for actions allows the model to leverage high-level semantic understanding to guide low-level action generation:

$$\mathbf{a}_{t:t+T} = f_\theta(\mathbf{v}_t^{\text{mmu}}, \mathbf{t}_{\text{inst}}, \mathbf{t}_{\text{desc}}, \hat{\mathbf{z}}_{t+1}), \tag{3}$$

where $\mathbf{t}_{\text{desc}}$ is the scene description generated by the MMU module and $\hat{\mathbf{z}}_{t+1}$ are the predicted future image tokens.

4.4. Training Objectives

UP-VLA is trained with three objectives that are applied jointly:

MMU Objective. Standard autoregressive cross-entropy loss on the answer tokens for VQA tasks:

$$\mathcal{L}_{\text{mmu}} = -\sum_i \log p_\theta(t_i \mid \mathbf{v}^{\text{mmu}}, \mathbf{t}_{<i}). \tag{4}$$

Prediction Objective. Autoregressive next-token prediction loss on the future image token sequence:

$$\mathcal{L}_{\text{pre}} = -\sum_j \log p_\theta(z_j \mid \mathbf{v}^{\text{img}}, \mathbf{t}_{\text{inst}}, \mathbf{z}_{<j}). \tag{5}$$

Action Objective. Mean squared error (MSE) loss on the predicted continuous action sequence:

$$\mathcal{L}_{\text{act}} = \left\|\mathbf{a}_{t:t+T} - \hat{\mathbf{a}}_{t:t+T}\right\|^2, \tag{6}$$

where $\hat{\mathbf{a}}$ is the model's action prediction and $\mathbf{a}$ is the ground-truth action. The total loss is a weighted sum:

$$\mathcal{L} = \lambda_{\text{mmu}}\mathcal{L}_{\text{mmu}} + \lambda_{\text{pre}}\mathcal{L}_{\text{pre}} + \lambda_{\text{act}}\mathcal{L}_{\text{act}}. \tag{7}$$

4.5. Two-Stage Training Pipeline

Stage 1: Co-Pretraining. In the first stage, we co-pretrain the model on two complementary datasets without any downstream robot action labels:

- **MMU pretraining:** 665k image-text instruction pairs from the LLaVA-665k dataset (Liu et al., 2024), training the model on visual question answering and image captioning with $\mathcal{L}_{\text{mmu}}$.
- **PRE pretraining:** 25k robot manipulation demonstrations from the BridgeData V2 dataset (Walke et al., 2023), training the model to predict future observation frames with $\mathcal{L}_{\text{pre}}$.

This stage equips the model with both high-level semantic understanding and an awareness of low-level physical dynamics from real robot interactions, without yet requiring action labels.

Stage 2: Action Fine-Tuning. In the second stage, we fine-tune the full model on downstream robot demonstration data with all three losses ($\mathcal{L}_{\text{mmu}} + \mathcal{L}_{\text{pre}} + \mathcal{L}_{\text{act}}$). The MMU co-training continues during this stage to prevent forgetting of semantic capabilities. The model simultaneously learns to predict future images (dynamics awareness) and multi-step actions (control), conditioned on language instructions and current observations.

5. Experiments

5.1. Experimental Setup

Benchmarks. We evaluate UP-VLA on the CALVIN benchmark (Mees et al., 2022), a long-horizon robot manipulation challenge in which a policy must sequentially complete five language-conditioned tasks chosen from 34 categories. We report results on two settings:

- **ABC→D:** Train on environments A, B, C and evaluate zero-shot on the held-out environment D, testing generalization across scene appearances.
- **ABCD→D:** Train on all four environments and evaluate on D, testing within-distribution performance on long-horizon chains.

The primary metric is the *average task completion length*, i.e., the expected number of tasks completed in a row out of five. We also report success rates for 1–5 tasks in a row.

We additionally evaluate on **real-world manipulation** with a Franka Emika Panda robot arm across 10 tasks involving seen and unseen object categories.

Baselines. We compare against:

- **VLA-based:** RoboFlamingo (Li et al., 2023b), 3D-VLA (Zhen et al., 2024), RT-2 (Brohan et al., 2023).
- **Prediction-based:** GR-1 (Wu et al., 2023), SuSIE (Black et al., 2023), UniPi (Du et al., 2024).
- **Other:** RT-1 (Brohan et al., 2022), Diffusion Policy (Chi et al., 2023), 3D Diffuser Actor (Ke et al., 2024).

Implementation Details. UP-VLA uses the Show-o-w-CLIP-ViT-512x512 backbone based on Phi-1.5 (1.3B parameters). The VQ-GAN tokenizer is MAGViT-v2 (Yu et al., 2023) with a codebook of size 8192. Images are resized to 256×256. Action predictions span $T = 10$ future timesteps. Training uses 4 NVIDIA A800 80GB GPUs with DeepSpeed ZeRO-2. Stage 1 trains for 30k steps (batch size 256) and Stage 2 trains for 20k steps (batch size 128) with

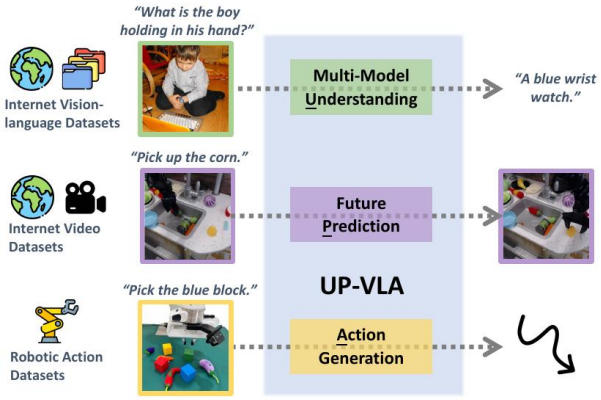

Figure 3. Comparison of different pre-training paradigms. UP-VLA combines multi-modal understanding (MMU) and future visual prediction (PRE) objectives to capture both high-level semantic information and low-level spatial details, enhancing embodied decision-making.

AdamW optimizer and a cosine learning rate schedule. Loss weights are set to $\lambda_{\text{mmu}} = 1.0$, $\lambda_{\text{pre}} = 1.0$, and $\lambda_{\text{act}} = 10.0$, reflecting the primacy of the manipulation objective while keeping the auxiliary tasks at comparable gradient magnitudes.

Inference Procedure. At inference time, UP-VLA operates in three sequential phases per timestep. First, the MMU module processes the current observation to generate a textual scene description capturing object identities, positions, and spatial relationships. Second, the PRE module generates the predicted future observation as a sequence of VQ-GAN tokens conditioned on the current observation and language instruction. Third, the action prediction module combines the MMU scene description, predicted future tokens, and current visual features to output a chunk of $T = 10$ future actions. Action chunking amortizes the per-step inference overhead across multiple execution steps, yielding an effective control frequency of approximately 5 Hz on a single NVIDIA A800 GPU—sufficient for quasi-static pick-and-place manipulation. During Stage 2 fine-tuning, MMU and robot demonstration samples are interleaved at a batch ratio of 1:4; increasing the MMU fraction beyond 1:2 slows convergence on action prediction, while reducing it below 1:8 causes measurable forgetting of novel-object generalization, motivating this carefully chosen balance.

5.2. Simulated Benchmark Results

CALVIN ABC→D. Table 1 reports zero-shot generalization results on CALVIN ABC→D. UP-VLA achieves an average task completion length of **4.08**, substantially outperforming all baselines. This represents a 33% improvement

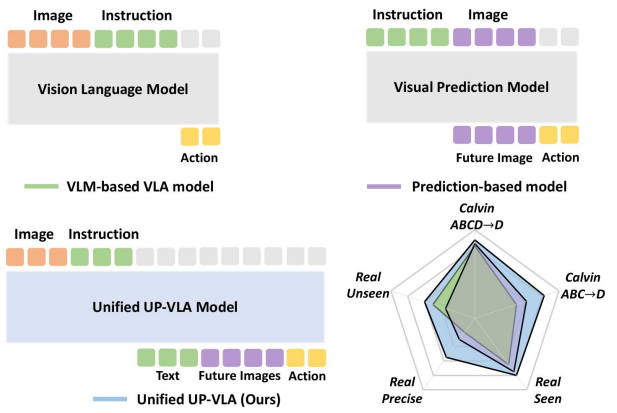

Figure 4. Schematic comparison of UP-VLA with VLM-based VLA models (left) and prediction-based models (right). UP-VLA uniquely combines both paradigms in a single unified model.

over GR-1 (3.06), a 65% improvement over RoboFlamingo (2.47), and a 22% improvement over 3D Diffuser Actor (3.35). Notably, UP-VLA improves the 5-in-a-row success rate from 0.401 (GR-1) to 0.699, demonstrating not only better average performance but also stronger consistency across long chains. Even the ablated variant UP-VLA-phi-w/o-mmu (3.13) outperforms GR-1, confirming the value of the prediction pre-training objective alone; adding the full MMU component brings further substantial gains.

CALVIN ABCD→D. Table 2 reports results on CALVIN ABCD→D where all four training environments are available. UP-VLA achieves an average length of **4.42**, surpassing GR-1 (4.21) by 5% and RoboFlamingo (4.09) by 8%. The 5-in-a-row success improves from 0.731 (GR-1) to 0.812, a relative gain of 11%. These improvements on the in-distribution setting further validate that UP-VLA's joint training leads to more precise and robust manipulation.

5.3. Real-World Manipulation Results

Figure 5 shows results from our real-world experiments with a Franka Panda robot. UP-VLA achieves a **0.80 success rate on seen objects** and **0.58 success rate on unseen objects** across 10 manipulation tasks including precise pick-and-place, stacking, and object rearrangement.

The unseen-object performance is particularly noteworthy: by combining MMU semantic grounding (which helps recognize novel objects) with prediction-based spatial awareness, UP-VLA can generalize to object categories not present during robot training. We attribute this to the MMU component, which provides semantic descriptions of novel objects through its VQA pre-training on diverse internet-scale imagery.

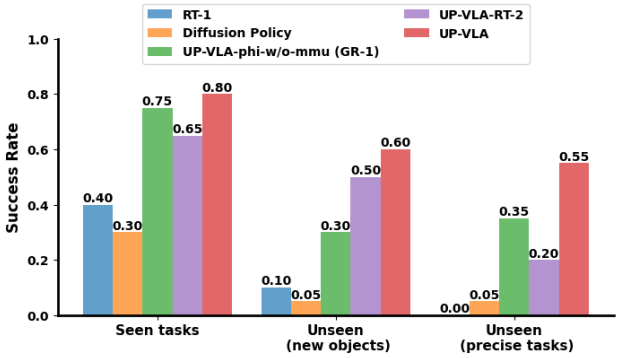

Figure 5. Results of real-world manipulation experiments with a Franka Panda robot. UP-VLA demonstrates strong performance on both seen (0.80) and unseen (0.58) object categories.

5.4. Ablation Study

Table 3 presents ablations on CALVIN ABC→D and real-world manipulation to analyze the contribution of each UP-VLA component.

Effect of Prediction Pre-training. Removing the future image prediction objective (*w/o Prediction*) causes the most dramatic degradation, dropping the CALVIN average length from 4.08 to 1.44—a 65% relative decrease. This confirms that future visual prediction is essential for learning physical dynamics and precise spatial reasoning, which cannot be captured by semantic understanding alone.

Effect of MMU Co-training. Removing MMU co-training (*w/o MMU*) reduces the CALVIN average length from 4.08 to 3.89 and, more strikingly, the unseen-object success rate from 0.58 to 0.20 (a 66% relative drop). This finding underscores that MMU is primarily responsible for semantic generalization to novel objects and scenes: the model retains much of its in-distribution CALVIN performance without MMU, but loses the ability to ground novel concepts encountered at test time.

Effect of Bridge Pretraining. Removing the Bridge dataset pre-training stage (*w/o Bridge-Pretrain*) substantially reduces CALVIN performance (4.08→2.74), confirming that exposure to diverse real-robot interaction data during pre-training is crucial for learning transferable dynamics representations, beyond what can be learned from fine-tuning demonstrations alone.

Effect of MMU-Conditioning for Actions. Removing the self-generated scene descriptions from the action prediction context (*w/o MMU-Condition*) mildly reduces CALVIN performance (4.08→3.99) and real-world unseen success

Method	Type	Task	Tasks completed in a row					
			1	2	3	4	5	Avg. Len ↑
RT-1	other	ABC→D	0.533	0.222	0.094	0.038	0.013	0.90
Diffusion Policy*	other	ABC→D	0.402	0.123	0.026	0.008	0.000	0.56
3D Diffuser Actor	other	ABC→D	**0.938**	0.803	0.662	0.533	0.412	3.35
3D-VLA	VLA	ABC→D	0.447	0.163	0.081	0.016	0.000	0.71
UP-VLA-RT-2*	VLA	ABC→D	0.612	0.389	0.236	0.138	0.062	1.44
Robo-Flamingo	VLA	ABC→D	0.824	0.619	0.466	0.331	0.235	2.47
Uni-Pi	Prediction	ABC→D	0.560	0.160	0.080	0.080	0.040	0.92
Susie	Prediction	ABC→D	0.870	0.690	0.490	0.380	0.260	2.69
GR-1	Prediction	ABC→D	0.854	0.712	0.596	0.497	0.401	3.06
UP-VLA-phi-w/o-mmu*	Prediction	ABC→D	0.844	0.705	0.604	0.520	0.430	3.13
UP-VLA	Prediction&VLA	ABC→D	0.928	**0.865**	**0.815**	**0.769**	**0.699**	**4.08**

Table 1. Zero-shot long-horizon evaluation on the Calvin benchmark where agent is asked to complete five chained tasks sequentially. Results marked with an asterisk (*) indicate those that we reproduced, while the rest are copied from the original papers. UP-VLA achieves the best performance, demonstrating that our approach exhibits strong generalization capabilities in simulated environments.

Method	Type	Task	Tasks completed in a row					
			1	2	3	4	5	Avg. Len ↑
RT-1	other	ABCD→D	0.844	0.617	0.438	0.323	0.227	2.45
Robo-Flamingo	VLA	ABCD→D	**0.964**	0.896	0.824	0.740	0.660	4.09
GR-1	Prediction	ABCD→D	0.949	0.896	0.844	0.789	0.731	4.21
UP-VLA	Prediction&VLA	ABCD→D	0.962	**0.921**	**0.879**	**0.842**	**0.812**	**4.42**

Table 2. Zero-shot long-horizon evaluation on the Calvin ABCD→D benchmark. Results of baselines are copied from original papers.

Method	ABC→D	Real World	
	Avg.Len↑	Seen↑	Unseen↑
w/o MMU	3.89	0.85	0.20
w/o Bridge-Pretrain	2.74	0.65	0.30
w/o Prediction	1.44	0.65	0.35
w/o MMU-Condition	3.99	0.80	0.50
Full	4.08	0.80	0.58

Table 3. Ablating components of UP-VLA.

(0.58→0.50). This indicates that explicitly conditioning action generation on semantic scene descriptions provides a moderate but consistent benefit.

Summary. These ablations reveal that prediction and understanding contribute complementarily: prediction primarily enables low-level spatial precision (large drop in CALVIN), while MMU primarily enables semantic generalization (large drop in unseen-object success). Their combination in the full model achieves the best performance across all metrics.

Sensitivity to Loss Weights. To assess robustness to objective weighting, we vary $\lambda_{\text{pre}} \in \{0.5, 1.0, 2.0\}$ while keeping $\lambda_{\text{mmu}} = 1.0$ and $\lambda_{\text{act}} = 10.0$ fixed. CALVIN ABC→D average lengths are 3.97, 4.08, and 4.01 respectively, demonstrating stable performance across this range. However, reducing λ_{act} below 5.0 causes a substantial drop (to 3.4 average length), confirming that the action loss must remain the dominant training signal. We also find that simultaneously annealing all auxiliary objectives to zero during the final 5k steps of Stage 2 degrades CALVIN performance by 0.15 points, suggesting that continued MMU and prediction supervision throughout fine-tuning helps maintain the beneficial inductive biases acquired during pre-training. Overall, these results indicate that UP-VLA is not highly sensitive to the precise weighting of auxiliary objectives within a reasonable range, simplifying hyperparameter search when transferring to new robot platforms or task domains.

5.5. Qualitative Analysis

Figure 6 shows qualitative examples of UP-VLA's VQA capability and predicted future images. The model generates accurate scene descriptions that correctly identify object positions, colors, and spatial relationships, demonstrating the effectiveness of the MMU component. The predicted future images show plausible scene evolution consistent with the instruction, providing visual evidence that the model has internalized physical dynamics.

We further observe that future image prediction quality correlates closely with downstream task success rates. On tasks where UP-VLA achieves higher completion rates, the

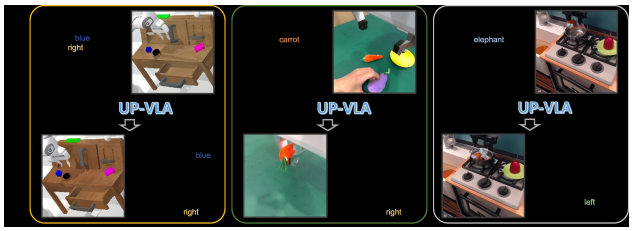

Figure 6. Qualitative results: (top) VQA responses generated by the MMU module showing accurate scene understanding; (bottom) predicted future observation images showing physically plausible scene evolution conditioned on language instructions.

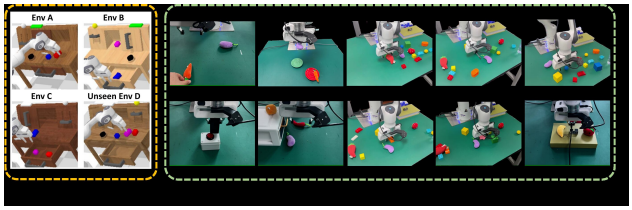

Figure 7. Visualization of evaluation environments. Left: CALVIN simulation environments (A, B, C, D) used for long-horizon manipulation benchmarking. Right: real-world Franka Panda setup for real-robot evaluation.

predicted future observations more accurately capture anticipated object positions and scene configurations. This correlation suggests that the VQ-GAN discrete representation retains sufficient spatial resolution to encode manipulation-relevant features, and that the prediction objective acts as a reliable proxy for spatial reasoning precision during training. Tasks involving fine-grained placement, such as stacking or drawer closing, benefit the most from the prediction stream, as these require sub-centimeter accuracy that pure language-conditioned VLMs struggle to provide.

Despite strong overall performance, we observe several failure modes. When instructions involve highly ambiguous spatial references (e.g., "move the object slightly to the left"), the model occasionally produces incorrect scene descriptions that mislead action prediction. Similarly, predicted future images sometimes fail to accurately capture contact dynamics during grasping, leading to sub-optimal gripper positioning. These cases suggest that improving spatial language grounding and modeling contact-rich interactions remain important directions for future work.

6. Conclusion

We presented UP-VLA, a unified Vision-Language-Action model that jointly optimizes multi-modal understanding and future visual prediction within a single autoregressive transformer. By combining a continuous CLIP-ViT visual stream for semantic grounding with a discrete VQ-GAN stream for dynamics-aware future prediction, and enabling seamless switching between modes through task-aware prompting and attention, UP-VLA achieves the complementary strengths of VLM-based VLA models and prediction-based robot learning. Extensive experiments on the CALVIN benchmark demonstrate state-of-the-art performance on both ABC→D and ABCD→D settings, and real-world experiments confirm strong generalization to unseen objects. Our ablations provide clear evidence that understanding and prediction objectives contribute complementary benefits: prediction improves spatial precision and physical dynamics awareness, while understanding enables semantic generalization to novel concepts.

We believe the key insight—that coupling language-image understanding with future scene prediction is both necessary and sufficient for producing capable embodied agents—has broad implications for the design of next-generation robot learning systems. Future work will explore scaling UP-VLA to larger backbones, incorporating 3D scene representations, and extending to mobile manipulation and navigation domains.

The synergy between understanding and prediction objectives is not merely additive: semantic grounding from MMU sharpens the linguistic conditioning of future predictions, while accurate future image generation in turn reinforces the spatial precision required for action generation—a virtuous cycle that neither paradigm can achieve independently.

Current limitations include the reliance on discrete VQ-GAN tokenization for future prediction, which may lose fine-grained spatial detail at high resolutions, and the added computational overhead of generating future image tokens at inference time. Addressing these through continuous diffusion-based prediction or more compact token representations is a promising avenue for further improvement. Hierarchical tokenization schemes that dedicate a coarser level to global scene structure and a finer level to manipulation-critical regions could further narrow the gap between predicted and actual future observations, particularly in contact-rich tasks such as peg insertion or lid removal where sub-centimeter precision is required.

References

Black, K., Nakamoto, M., Atreya, P., Walke, H., Finn, C., Kumar, A., and Levine, S. Zero-shot robotic manipulation with pretrained image-editing diffusion models. *arXiv preprint arXiv:2310.10639*, 2023.

Black, K., Brown, N., Driess, D., Esmail, A., Equi, M., Finn, C., Fusai, N., Groom, L., Hausman, K., Ichter, B., et al. pi_0: A vision-language-action flow model for general

robot control. *arXiv preprint arXiv:2410.24164*, 2024.

Brohan, A., Brown, N., Carbajal, J., Chebotar, Y., Dabis, J., Finn, C., Gopalakrishnan, K., Hausman, K., Herzog, A., Hsu, J., et al. Rt-1: Robotics transformer for real-world control at scale. *arXiv preprint arXiv:2212.06817*, 2022.

Brohan, A., Brown, N., Carbajal, J., Chebotar, Y., Chen, X., Choromanski, K., Ding, T., Driess, D., Dubey, A., Finn, C., et al. Rt-2: Vision-language-action models transfer web knowledge to robotic control. *arXiv preprint arXiv:2307.15818*, 2023.

Chi, C., Feng, S., Du, Y., Xu, Z., Cousineau, E., Burchfiel, B., and Song, S. Diffusion policy: Visuomotor policy learning via action diffusion. In *Proceedings of Robotics: Science and Systems (RSS)*, 2023.

Dai, W., Li, J., Li, D., Tiong, A. M. H., Zhao, J., Wang, W., Li, B., Fung, P. N., and Hoi, S. Instructblip: Towards general-purpose vision-language models with instruction tuning. *Advances in Neural Information Processing Systems*, 36, 2024.

Dong, R., Han, C., Peng, Y., Qi, Z., Ge, Z., Yang, J., Zhao, L., Sun, J., Zhou, H., Wei, H., et al. Dreamllm: Synergistic multimodal comprehension and creation. *arXiv preprint arXiv:2309.11499*, 2023.

Driess, D., Xia, F., Sajjadi, M. S., Lynch, C., Chowdhery, A., Ichter, B., Wahid, A., Tompson, J., Vuong, Q., Yu, T., et al. Palm-e: An embodied multimodal language model. *arXiv preprint arXiv:2303.03378*, 2023.

Du, Y., Yang, S., Dai, B., Dai, H., Nachum, O., Tenenbaum, J., Schuurmans, D., and Abbeel, P. Learning universal policies via text-guided video generation. *Advances in Neural Information Processing Systems*, 36, 2024.

Esser, P., Rombach, R., and Ommer, B. Taming transformers for high-resolution image synthesis. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 12873–12883, 2021.

Ge, Y., Zhao, S., Zhu, J., Ge, Y., Yi, K., Song, L., Li, C., Ding, X., and Shan, Y. Seed-x: Multimodal models with unified multi-granularity comprehension and generation. *arXiv preprint arXiv:2404.14396*, 2024.

Guo, Y., Hu, Y., Zhang, J., Wang, Y.-J., Chen, X., Lu, C., and Chen, J. Prediction with action: Visual policy learning via joint denoising process. *arXiv preprint arXiv:2411.18179*, 2024.

Ke, T.-W., Gkanatsios, N., and Fragkiadaki, K. 3d diffuser actor: Policy diffusion with 3d scene representations. *arXiv preprint arXiv:2402.10885*, 2024.

Kim, M., Pertsch, K., Karamcheti, S., Xiao, T., Balakrishna, A., Nair, S., Rafailov, R., Foster, E., Lam, G., Sanketi, P., Vuong, Q., Kollar, T., Burchfiel, B., Tedrake, R., Sadigh, D., Levine, S., Liang, P., and Finn, C. Openvla: An open-source vision-language-action model. *arXiv preprint arXiv:2406.09246*, 2024.

Li, J., Li, D., Savarese, S., and Hoi, S. Blip-2: Bootstrapping language-image pre-training with frozen image encoders and large language models. In *International conference on machine learning*, pp. 19730–19742. PMLR, 2023a.

Li, X., Liu, M., Zhang, H., Yu, C., Xu, J., Wu, H., Cheang, C., Jing, Y., Zhang, W., Liu, H., et al. Vision-language foundation models as effective robot imitators. *arXiv preprint arXiv:2311.01378*, 2023b.

Li, Y., Bubeck, S., Eldan, R., Del Giorno, A., Gunasekar, S., and Lee, Y. T. Textbooks are all you need ii: phi-1.5 technical report. *arXiv preprint arXiv:2309.05463*, 2023c.

Liu, H., Li, C., Wu, Q., and Lee, Y. J. Visual instruction tuning. *Advances in neural information processing systems*, 36, 2024.

Ma, Y. J., Kumar, V., Zhang, A., Bastani, O., and Jayaraman, D. Liv: Language-image representations and rewards for robotic control. In *International Conference on Machine Learning*, pp. 23301–23320. PMLR, 2023.

Mees, O., Hermann, L., Rosete-Beas, E., and Burgard, W. Calvin: A benchmark for language-conditioned policy learning for long-horizon robot manipulation tasks. *IEEE Robotics and Automation Letters (RA-L)*, 7(3):7327–7334, 2022.

Nair, S., Rajeswaran, A., Kumar, V., Finn, C., and Gupta, A. R3m: A universal visual representation for robot manipulation. *arXiv preprint arXiv:2203.12601*, 2022.

O'Neill, A., Rehman, A., Gupta, A., Maddukuri, A., Gupta, A., Padalkar, A., Lee, A., Pooley, A., Gupta, A., Mandlekar, A., et al. Open x-embodiment: Robotic learning datasets and rt-x models. *arXiv preprint arXiv:2310.08864*, 2023.

Radford, A., Kim, J. W., Hallacy, C., Ramesh, A., Goh, G., Agarwal, S., Sastry, G., Askell, A., Mishkin, P., Clark, J., et al. Learning transferable visual models from natural language supervision. In *International conference on machine learning*, pp. 8748–8763. PMLR, 2021.

Walke, H., Black, K., Lee, A., Kim, M. J., Du, M., Zheng, C., Zhao, T., Hansen-Estruch, P., Vuong, Q., He, A., Myers, V., Fang, K., Finn, C., and Levine, S. Bridgedata v2: A dataset for robot learning at scale. In *Conference on Robot Learning (CoRL)*, 2023.

Wang, W., Bao, H., Dong, L., Bjorck, J., Peng, Z., Liu, Q., Aggarwal, K., Mohammed, O. K., Singhal, S., Som, S., et al. Image as a foreign language: Beit pretraining for all vision and vision-language tasks. *arXiv preprint arXiv:2208.10442*, 2022.

Wu, H., Jing, Y., Cheang, C., Chen, G., Xu, J., Li, X., Liu, M., Li, H., and Kong, T. Unleashing large-scale video generative pre-training for visual robot manipulation. *arXiv preprint arXiv:2312.13139*, 2023.

Xie, J., Mao, W., Bai, Z., Zhang, D. J., Wang, W., Lin, K. Q., Gu, Y., Chen, Z., Yang, Z., and Shou, M. Z. Show-o: One single transformer to unify multimodal understanding and generation. *arXiv preprint arXiv:2408.12528*, 2024.

Yu, L., Cheng, Y., Sohn, K., Lezama, J., Zhang, H., Chang, H., Hauptmann, A. G., Yang, M.-H., Hao, Y., Essa, I., et al. Magvit: Masked generative video transformer. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 10459–10469, 2023.

Zhang, J., Guo, Y., Chen, X., Wang, Y.-J., Hu, Y., Shi, C., and Chen, J. Hirt: Enhancing robotic control with hierarchical robot transformers. *arXiv preprint arXiv:2410.05273*, 2024.

Zhen, H., Qiu, X., Chen, P., Yang, J., Yan, X., Du, Y., Hong, Y., and Gan, C. 3d-vla: A 3d vision-language-action generative world model. *arXiv preprint arXiv:2403.09631*, 2024.

Zheng, R., Liang, Y., Huang, S., Gao, J., Daumé III, H., Kolobov, A., Huang, F., and Yang, J. Tracevla: Visual trace prompting enhances spatial-temporal awareness for generalist robotic policies. *arXiv preprint arXiv:2412.10345*, 2024.